

SkyTrace: Missing Aircraft Search Area Estimator

Helping narrow down a possible aircraft search area.

Last contact details

Last known latitude

21.01106

Last known longitude

79.76074

Or click anywhere on the map to set the last known position.

Time since last contact (minutes)

39

Aircraft speed (km/h)

589

Heading (0-359°)

287

287

0 = north, 90 = east, 180 = south, 270 = west

Uncertainty setting

Moderate

An illustrative modelling assumption about how widely the area spreads – not a measured confidence level.

Show search grid with demo priorities

Time since contact

—

Estimated travel distance

—

Outer search area

—

Fit map to search area

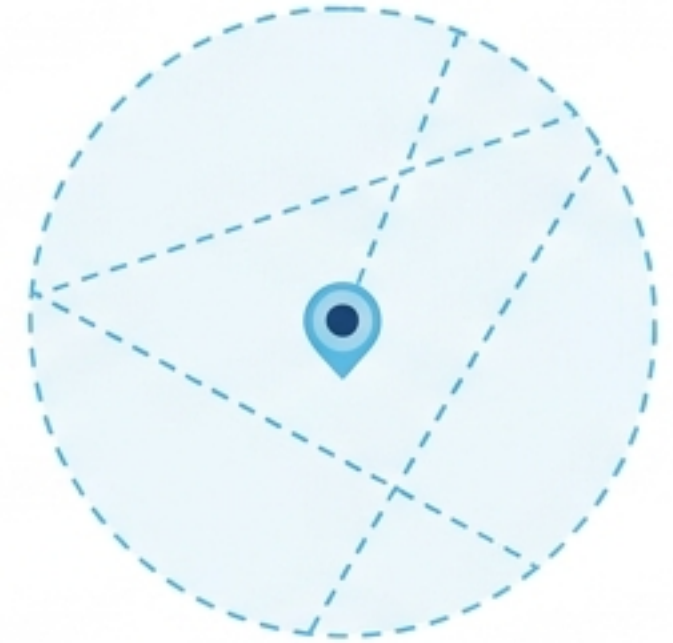
● Last known position
● Estimated central position
● Inner uncertainty zone
● Outer uncertainty zone
— Estimated path (dashed)
 Grid: red High - amber Medium - green Low (demo)

[Leaffel](#) | [Tiles & Earth](#) | [Gemini Notebook](#)

Searching Everywhere Wastes Valuable Time and Resources

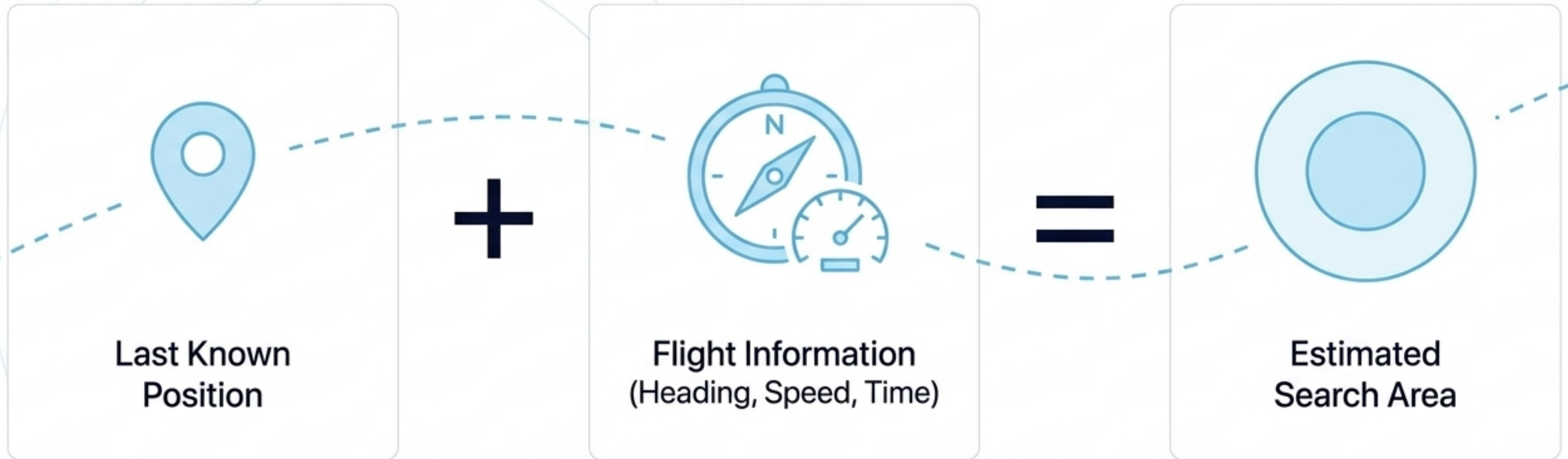


When an aircraft goes missing, search teams are often left with a massive geographic area to investigate.



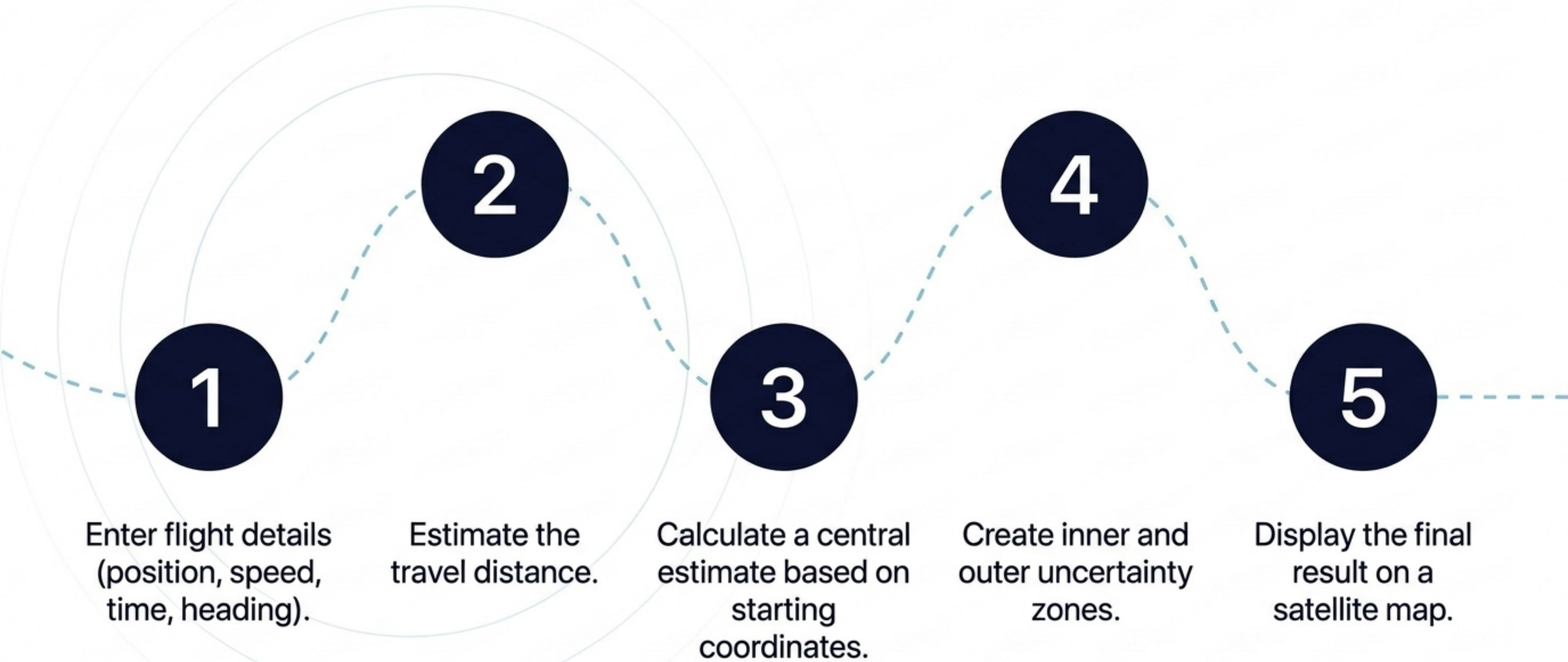
Without a way to narrow down the possibilities, searching everywhere wastes critical time and resources.

Estimating a Targeted Search Area from Flight Data

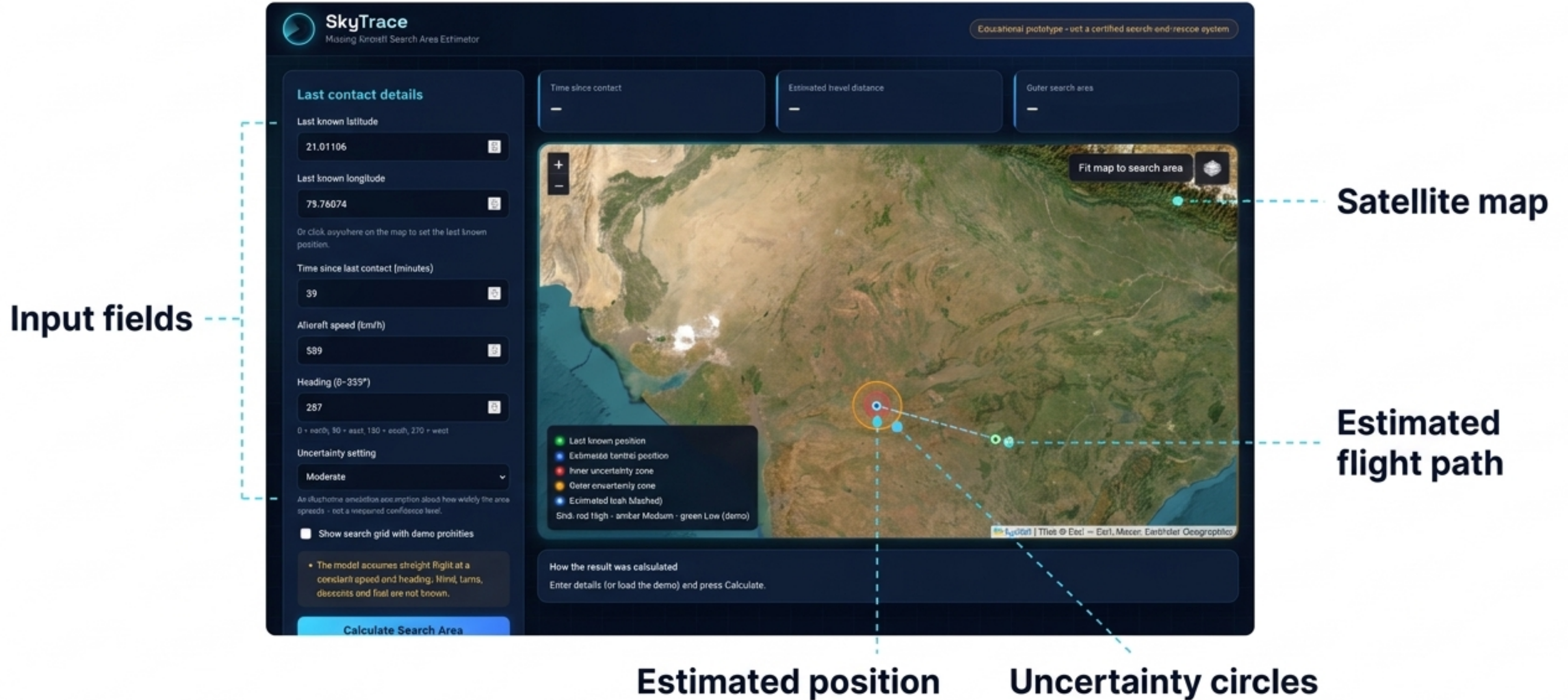


SkyTrace is a web application that takes the last known coordinates and available flight details to calculate and display a targeted region on a map, rather than just a single point.

A Simple Five-Step Calculation Workflow



The SkyTrace Interface



Current Capabilities vs. Future Vision

Current Features

- ✓ Flight information input.
- ✓ Distance and position estimation.
- ✓ Satellite map and uncertainty zones.
- ✓ Optional demonstration priority grid.

Future Improvements

- Weather and wind information.
- More realistic aircraft movement estimates.
- Better search-priority methods.

Turning Flight Information into Visual Estimates

SkyTrace successfully demonstrates how basic flight data can be transformed into a visual map, moving search logic from abstract coordinates to a defined possible search area.

- SkyTrace is strictly an educational prototype, not a certified search-and-rescue system.
- The uncertainty zones are illustrative modelling assumptions, not verified probabilities.
 - The prototype does not use AI or live weather data.

Meet Team ERROR 404

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Thank You.